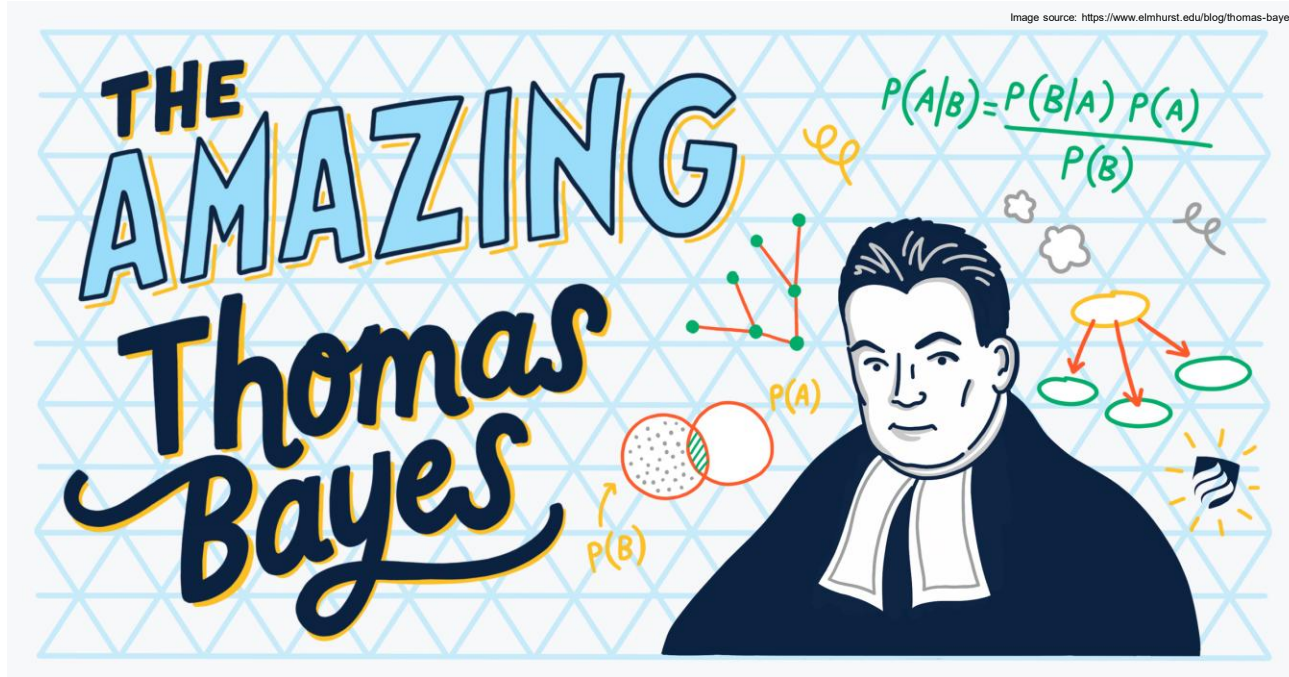


Naïve Bayes Classifier (NB)



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What is Naïve Bayes (NB)?

- NB is a **classification algorithm**, like k-NN, but conceptually different
 - NB is often used to in text analysis, i.e., classify an incoming email as “spam” vs “not spam” (ham)
- Classification algorithms have one main objective:
 - To compute the probability that a new data point belong to a class (e.g., y =“spam”), given a set of features describing it (x), meaning, we are trying to compute the conditional probability $P(y=\text{“spam”} \mid \text{the email contains term } x=\text{“free money”})$

There are 2 main types of Naïve Bayes Algos

- Multinomial NB (MNB)

- Best when all features/inputs (X) are categorical
 - Example #1: “Education level” belongs to either “BS” “MS” or PhD”
 - Example #2: “Customer already has a credit card account” is either “1 or 0”
- If a few features are numerical... you can try “binning” to convert to categorical
 - Example: “Customer Age” can be binned into categories “0 to 17”, “18-29”, “30-44”, etc.

- Gaussian NB (GNB)

- When most inputs (X) are numerical, and are drawn from an identifiable (continuous) distribution
- Gaussian means we assume values from a normal distribution
- Example: “Yearly Income” ranges from [0,200], with a mean of 100, and stdev of 50

- MNB is arguably more successful (makes less assumptions) so we will focus on this...

Reminder on Bayes' Theorem

The diagram illustrates Bayes' Theorem with the following components:

- LIKELIHOOD** (orange text): the probability of "B" being TRUE given that "A" is TRUE. An arrow points from this text to the orange box containing $P(B|A)$.
- PRIOR** (teal text): the probability of "A" being TRUE. An arrow points from this text to the teal box containing $P(A)$.
- POSTERIOR** (green text): the probability of "A" being TRUE given that "B" is TRUE. An arrow points from this text to the green box containing $P(A|B)$.
- Marginal Likelihood** (pink text): The probability of "B" being TRUE. An arrow points from this text to the pink box containing $P(B)$.

The equation is presented as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

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also remember, the "law of total probability":

$$P(B) = P(B|A)P(A) + P(B|\text{not } A)P(\text{not } A)$$

Example

- Mini dataset of firms' tax return audits
- Prediction problem: classify a new firm as either “fraudulent” or “truthful”
- 10 past observations, 2 features in X :
 - Has the firm faced passed charges? “yes” or “no”
 - What is the size of the firm? “small” or “large”
- New firm has a feature vector:
 $x = \{\text{Charges} = \text{“yes”}, \text{Size} = \text{“small”}\}$
- Is the new firm fraudulent?
 - What's the probability $P(\text{fraud} \mid \text{yes}, \text{small})$?
 - Idea: let's look at other such similar firms...

X		y
Charges?	Size	Outcome
yes	small	truthful
no	small	truthful
no	large	truthful
no	large	truthful
no	small	truthful
no	small	truthful
yes	small	fraud
yes	small	fraud
no	large	fraud
yes	large	fraud

Example (direct) discriminative approach

- How many firms (rows) have the exact same features:

$x = \{\text{"yes"}, \text{"small"}\}?$

- Answer: 3 total observations... (in red)
 - One outcome "truthful" but the other 2 are "fraud"
 - $P(y=\text{fraud} \mid \text{"yes"}, \text{"small"}) = 2/3 = 0.67$
- So it seems like the new firm has a high chance of being fraudulent...

X		Y
Charges?	Size	Outcome
yes	small	truthful
no	small	truthful
no	large	truthful
no	large	truthful
no	small	truthful
no	small	truthful
yes	small	fraud
yes	small	fraud
no	large	fraud
yes	large	fraud

Example (indirect) generative approach

- Bayes flips order around.
 - Simplify notation (small=s, fraud=f, yes=y, truthful=t)
 - $P(y,s | f) = 2/4 = 1/2$
 - $P(f) = 4/10$
 - $P(y,s | t) = 1/6$
 - $P(t) = 6/10$
 - $P(y,s) = P(y,s | f) P(f) + P(y,s | t) P(t)$ (law of Tot. Prob.)
- All together now:

$$P(f|y,s) = \frac{P(y,s|f)P(f)}{P(y,s|f)P(f) + P(y,s|t)P(t)}$$

$$P(f|y,s) = \frac{\frac{1}{2} * \frac{4}{10}}{\frac{1}{2} * \frac{4}{10} + \frac{1}{6} * \frac{6}{10}} = \frac{\frac{2}{10}}{\frac{2}{10} + \frac{1}{10}} = \frac{2}{3} = 0.67!$$

X		Y
Charges?	Size	Outcome
yes	small	truthful
no	small	truthful
no	large	truthful
no	large	truthful
no	small	truthful
no	small	truthful
yes	small	fraud
yes	small	fraud
no	large	fraud
yes	large	fraud

- Great, both discriminative and generative methods give the same result...
 - But what if we can't find an "exact" match?

The Naïve Bayes approach

- When the number of features increases, finding an exact match (direct approach) becomes increasingly difficult
- Naïve Bayes then steps in (“like a boss”) and assumes independence between all the features!

$$P(\text{yes, small}|\text{fraud}) = P(\text{yes}|\text{fraud}) * P(\text{small}|\text{fraud}) !!$$

- Even if we don’t happen to see any exact “yes,small” cases, we do independently observe “yes” cases and “small” cases
- How can this possibly work??
 - We will get wrong probabilities, but... we are more interested in something else: the “rank order” between the outcomes, and this, is preserved! (see chap 8, of DMBA book)

X		y
Charges?	Size	Outcome
yes	small	truthful
no	small	truthful
no	large	truthful
no	large	truthful
no	small	truthful
no	small	truthful
yes	small	fraud
yes	small	fraud
no	large	fraud
yes	large	fraud

Let's now revisit with the Naïve Bayes assumption...

- Assuming independence, we now need to compute:

- $P(y | f) = 3/4$
- $P(s | f) = 2/4 = 1/2$
- $P(y | t) = 1/6$
- $P(s | t) = 4/6$
- $P(f) = 4/10$
- $P(t) = 6/10$

- All together now:

$$P(f|y, s) \triangleq \frac{P(y | f)P(s|f)P(f)}{P(y | f)P(s|f)P(f) + P(y | t)P(s|t)P(t)}$$

$$P(f|y, s) = \frac{\frac{3}{4} * \frac{1}{2} * \frac{4}{10}}{\frac{3}{4} * \frac{1}{2} * \frac{4}{10} + \frac{1}{6} * \frac{4}{6} * \frac{6}{10}} = \dots = 0.6923$$

- Very close to true probability $P(\text{fraud} | \text{yes}, \text{small}) = 0.67$! We should (correctly) classify the new firm as fraudulent.

X		y
Charges?	Size	Outcome
yes	small	truthful
no	small	truthful
no	large	truthful
no	large	truthful
no	small	truthful
no	small	truthful
yes	small	fraud
yes	small	fraud
no	large	fraud
yes	large	fraud

One last thing... regarding “Laplace Smoothing”

- Imagine if the dataset was limited and didn't cover all cases:

- $P(y | f) = 0$ (instead of $\frac{3}{4}$)
- $P(s | f) = 1/4$
- ...

$$P(f|y, s) \triangleq \frac{P(y|f)P(s|f)P(f)}{P(y|f)P(s|f)P(f) + P(y|t)P(s|t)P(t)}$$
$$P(f|y, s) = 0$$

- That doesn't really “feel” right
 - To rectify this “boundary” issue, we use something called a “Laplace smoothing” correction
 - Intuitively, it involves adding a few “fake entries” $\alpha > 0$ to the dataset, to remove the zero cases

$$replace\ P(y|f) = \frac{\#events}{\#total_observations}$$

$$with\ P(y|f) = \frac{\#events + \alpha}{\#total_observations + \alpha * \#dim}$$

If that felt a bit tedious...

You can thank:

Thomas Bayes
(1701-1761)



Pierre-Simon Laplace
(1749-1827)



We will use **python** to automate all of these calculations (“like a boss”)...